

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location RagingWire CA3, CA -- station KMEV, America/Los_Angeles
Building OSM 190868312
RagingWire CA3
Facade gap not applicable -- one building, no second facade
Weather record 42,111 real hourly records from KMEV
Plume physics NOT MODELLED. No other tagged data centre lies inside the solver's validated range, so there is no neighbour intake for a plume to arrive at: the quantity does not exist here rather than being unmeasured. This is a statement about the model's domain, not a claim that recirculation here is zero. A building's own exhaust re-entering its own intake is not modelled at any site in this project.

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2021-04-02 (crossing)
Plant limit 18.0 C
Notice required 3 h
Forecast skill 0.50 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 1 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 5 of 24 hours with 1 mode change. The reactive incumbent took 12 hours -- 7 more -- but declared 1 unsafe hour against the agent's 0. The incumbent had to break its own switch budget 1 time to stay safe; the agent never did. 11 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 5 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent 12 of 24 hours free cooling, 2 mode change(s), 1 unsafe hour(s)
the incumbent broke its own switch budget 1 time(s) to stay safe
11 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	yes	switch budget	0.000	18.0	0.000	1.632
01	mechanical	yes	switch budget	-0.500	18.0	-1.000	1.385
02	mechanical	yes	switch budget	-1.000	18.0	-2.000	1.233
03	mechanical	yes	switch budget	-0.500	18.0	-1.000	1.109
04	mechanical	yes	switch budget	-1.000	18.0	-1.000	1.008
05	mechanical	yes	switch budget	-3.000	18.0	-5.000	1.387

06	mechanical	yes	switch budget	-2.500	18.0	-7.000	1.712
07	mechanical	yes	switch budget	3.500	18.0	0.000	2.342
08	mechanical	yes	switch budget	7.500	18.0	7.000	2.889
09	mechanical	yes	switch budget	9.000	18.0	11.000	2.356
10	mechanical	yes	switch budget	14.500	18.0	16.000	2.216
11	mechanical	no	dry-bulb	19.000	18.0	20.000	1.962
12	mechanical	no	dry-bulb	21.000	18.0	21.000	2.130
13	mechanical	no	dry-bulb	23.500	18.0	23.000	1.823
14	mechanical	no	dry-bulb	25.000	18.0	24.000	1.569
15	mechanical	no	dry-bulb	24.500	18.0	24.000	1.376
16	mechanical	no	dry-bulb	23.000	18.0	21.000	1.443
17	mechanical	no	dry-bulb	22.000	18.0	20.000	1.552
18	mechanical	no	dry-bulb	21.000	18.0	19.000	1.829
19	FREE	yes	-	17.000	18.0	15.000	1.776
20	FREE	yes	-	14.000	18.0	9.000	2.311
21	FREE	yes	-	12.500	18.0	7.000	2.281
22	FREE	yes	-	10.500	18.0	7.000	1.941
23	FREE	yes	-	6.500	18.0	4.000	2.046

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 0.000 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

01:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -0.500 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

02:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -1.000 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

03:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -0.500 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

04:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -1.000 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is

also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -3.000 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is -2.500 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

07:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 3.500 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

08:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 7.500 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

09:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 9.000 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

10:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 14.500 C against a 18.0 C limit. The schedule is what forbids it: the switch budget of 1 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 19.000 C against a 18.0 C limit, so it fails by 1.000 C. It would take a limit 1.000 C higher, or a bound 1.000 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.000 C against a 18.0 C limit, so it fails by 3.000 C. It would take a limit 3.000 C higher, or a bound 3.000 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.500 C against a 18.0 C limit, so it fails by 5.500 C. It would take a limit 5.500 C higher, or a bound 5.500 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.000 C against a 18.0 C limit, so it fails by 7.000 C. It would take a limit 7.000 C higher, or a bound 7.000 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.500 C against a 18.0 C limit, so it fails by 6.500 C. It would take a limit 6.500 C higher, or a bound 6.500 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.000 C against a 18.0 C limit, so it fails by 5.000 C. It would take a limit 5.000 C higher, or a bound 5.000 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.000 C against a 18.0 C limit, so it fails by 4.000 C. It would take a limit 4.000 C higher, or a bound 4.000 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.000 C against a 18.0 C limit, so it fails by 3.000 C. It would take a limit 3.000 C higher, or a bound 3.000 C tighter, to change this hour.

19:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.000 C, which is 1.000 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.775 C, and the margin is measured, not chosen: 1.775 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

20:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 14.000 C, which is 4.000 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.311 C, and the margin is measured, not chosen: 2.311 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

21:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 12.500 C, which is 5.500 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.281 C, and the margin is measured, not chosen: 2.281 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

22:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 10.500 C, which is 7.500 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 1.941 C, and the margin is measured, not chosen: 1.941 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 6.500 C, which is 11.500 C under the 18.0 C plant limit. That bound is the forecast plus a margin of 2.046 C, and the margin is measured, not chosen: 2.046 C of group-conditional forecast error for this hour of day, plus 0.0000 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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